

Calculus Sheet 1

Differentiation and integration (synthetic sample)

Exercise 1. Compute the derivative of $f(x) = x^3 - 2x + 5$ and evaluate it at $x = 2$. Show every step of your computation.

Exercise 2. Let $g(x) = \sin(x) + x^2$ on the interval $[0, \pi]$. Answer the following questions.

- (a) Compute the derivative $g'(x)$.
- (b) Find all critical points of g in the interval $[0, \pi]$.
- (c) Decide for each critical point whether it is a local minimum or a local maximum, and justify your answer.

Instructions for Exercises 3 and 4

Justify every answer with a short proof. You may use the standard rules of differentiation without proving them again.

Exercise 3. Prove that $h(x) = |x|$ is not differentiable at $x = 0$.

Hint: compare the one-sided limits of the difference quotient.

Exercise 4. Let p be a polynomial of degree $n \geq 1$ with real coefficients and let $a < b$ be two real numbers with $p(a) = p(b) = 0$. Show that the derivative p' has at least one zero in the open interval

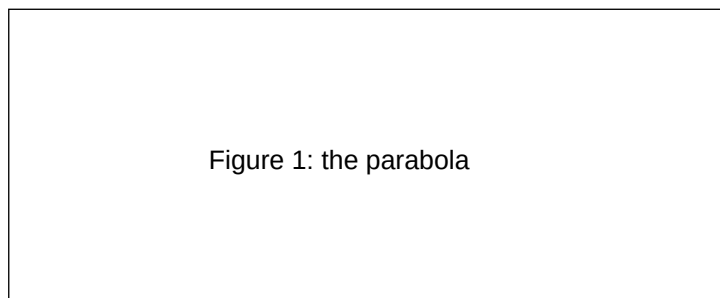
(a, b) , counted with multiplicity. State the theorem that you use.

Definition 5.1 (Derivative)

Let f be defined near a . If the limit of $(f(a + h) - f(a)) / h$ as h tends to 0 exists, it is called the derivative of f at a and is written $f'(a)$.

Exercise 5. Let $f(x) = x^2$ for x in $[0, 2]$.

- a) Sketch the graph of f and mark the tangent line at $x = 1$.
- b) Compute the area under the graph with an integral.



Remark. The tangent line at a point is the best linear approximation of the function near that point.